

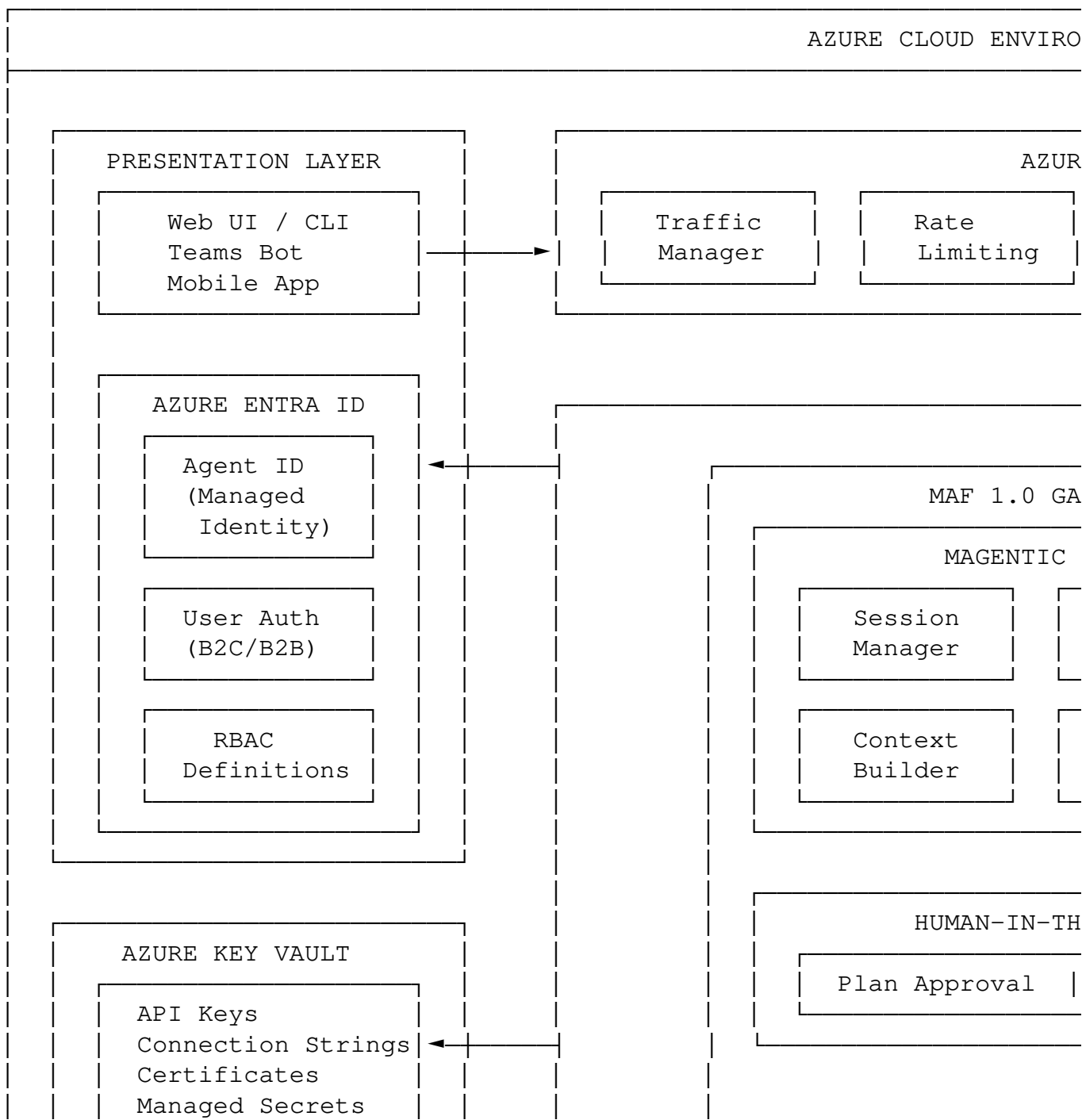
MAFGA Multi-Agent Architecture Diagrams

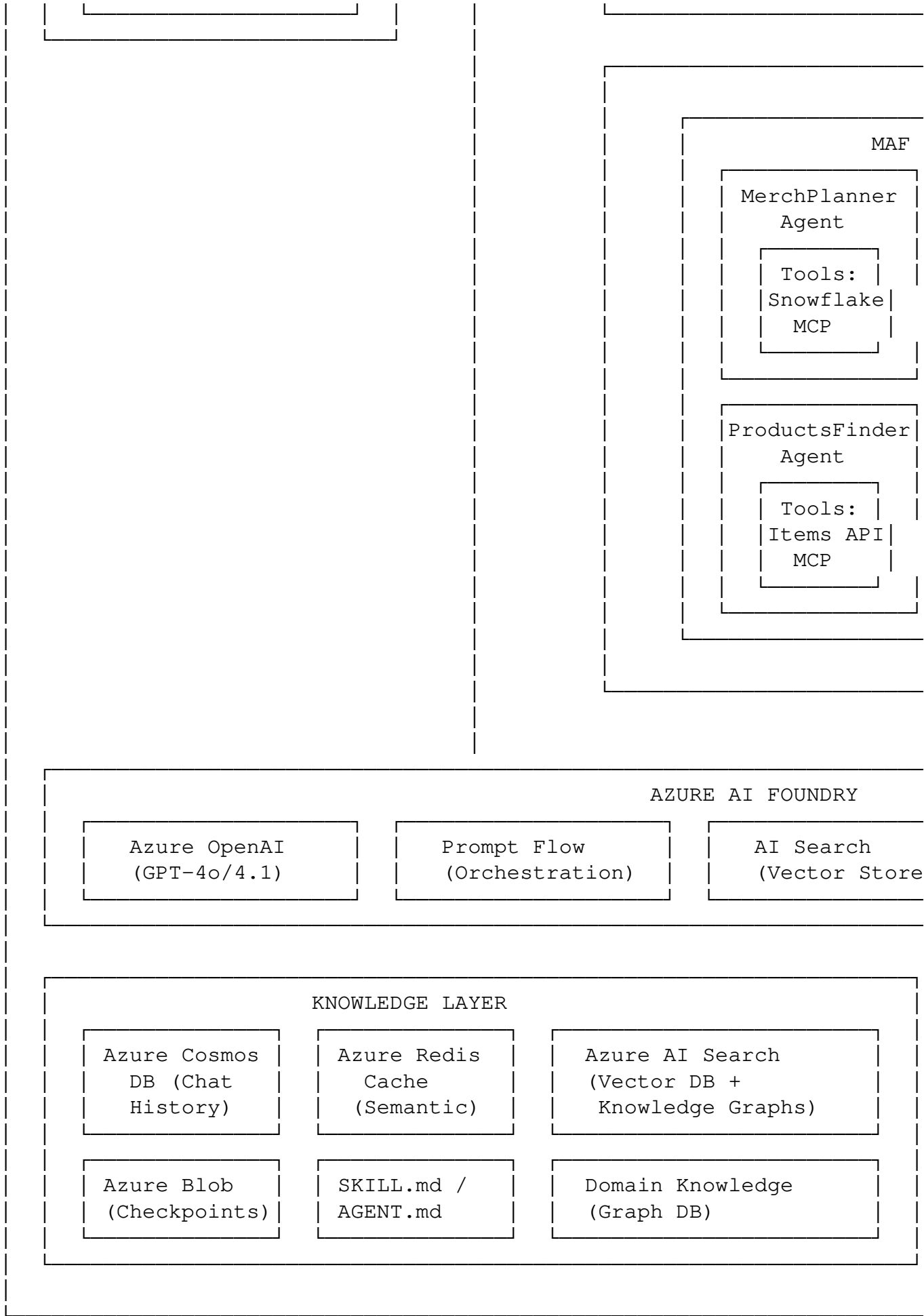
This document provides three versions of the architecture based on different deployment scenarios.

Version 1: Azure-Native (Foundry + APIM + Entra ID + KeyVault + MAF GA)

This version leverages the full Azure ecosystem for enterprise-grade deployment.

ASCII Diagram





Mermaid Diagram

flowchart TB

```
subgraph Clients["👤 Client Applications"]
    WebUI["🌐 Web UI"]
    Teams["💬 Teams Bot"]
    Mobile["📱 Mobile App"]
    CLI["🖥️ CLI"]
end
```

end

```
subgraph EntraID["🔑 Azure Entra ID"]
    AgentID["👤 Agent Managed Identity"]
    UserAuth["👤 User Authentication (B2C/B2B)"]
    RBAC["❤️ RBAC Definitions"]
    AppReg["📄 App Registrations"]
end
```

end

```
subgraph KeyVault["🔑 Azure Key Vault"]
    APIKeys["🔑 API Keys"]
    ConnStrings["🔑 Connection Strings"]
    Certs["📄 Certificates"]
    Secrets["🔒 Managed Secrets"]
end
```

end

```
subgraph APIM["🌐 Azure API Management"]
    Traffic["📶 Traffic Manager"]
    RateLimit["🕒 Rate Limiting"]
    Validation["✅ Request Validation"]
    OAuth["🔑 OAuth2 Gateway"]
    Policies["📄 Policy Engine"]
end
```

end

```
subgraph MAF_Orchestration["⚙️ MAF 1.0 GA - Orchestration Layer"]
    subgraph Magentic["🌀 Magentic Orchestrator"]
        SessionMgr["📄 Session Manager"]
        Planner["📅 LLM Planner"]
        Synthesizer["📝 Result Synthesizer"]
        ContextBuilder["🏗️ Context Builder"]
        AgentRouter["🔄 Agent Router"]
        Logger["📶 Logger"]
    end
end
```

end

```
subgraph HITL["👁️ Human-in-the-Loop"]
    Approval["✅ Plan Approval"]
    AsyncQueue["👤 Async Queue"]
    Timeout["🕒 Timeout Handler"]
end
```

end

```
TrafficRouter["🔄 Traffic Router (Handoff)"]
ParallelExec["⚡ Parallel Executor"]
```

end

```
subgraph MAF_Agents["👤 MAF 1.0 GA - Agent Layer"]
    MerchPlanner["📦 MerchPlanner Agent<br/>Tools: Snowflake MCP"]
    SpacePlanner["🏠 SpacePlanner Agent<br/>Tools: Snowflake MCP"]
end
```

```

    LoyaltyAgent["🛍️ LoyaltyAgent<br/>Tools: Personal, Localise MCP  

    ProductsFinder["🔍 ProductsFinder Agent<br/>Tools: Items API MCI  

    CommercialSales["📁 CommercialSales Agent<br/>Tools: Salesforce  

    CampaignAnalyst["📊 CampaignAnalyst Agent<br/>Tools: Weather, S  

end

subgraph Foundry[" Azure AI Foundry"]
    AzureOpenAI[" Azure OpenAI<br/>(GPT-4o / GPT-4.1)"]
    PromptFlow["🔄 Prompt Flow"]
    AISearch["🔍 AI Search"]
    ContentSafety["🛡️ Content Safety"]
    Models["📦 Model Deployments"]
end

subgraph Knowledge["📚 Knowledge Layer"]
    CosmosDB[("📄 Cosmos DB<br/>Chat History")]
    RedisCache[("🚀 Redis Cache<br/>Semantic Cache")]
    VectorDB[("📊 AI Search<br/>Vector Store")]
    BlobStorage[("📁 Blob Storage<br/>Checkpoints")]
    SkillsMD["📄 SKILL.md / AGENT.md"]
end

subgraph MCP_Layer["🔗 MCP / A2A Layer"]
    subgraph InternalMCP["Internal MCP"]
        SnowflakeMCP["❄️ Snowflake MCP"]
        SalesforceMCP["☁️ Salesforce MCP"]
        WeatherMCP["🌤️ Weather MCP"]
        ItemsMCP["📦 Items API MCP"]
    end
    subgraph ExternalA2A["External A2A"]
        PricingA2A["💰 Pricing Agent"]
        InventoryA2A["📊 Inventory Agent"]
        FulfilmentA2A["🚚 Fulfilment Agent"]
    end
end

subgraph Observability["📊 Observability"]
    AppInsights["📊 Application Insights"]
    LogAnalytics["🔍 Log Analytics"]
    Langfuse["📝 Langfuse (LLM Obs)"]
end

%% Connections
Clients --> APIM
APIM --> EntraID
EntraID --> MAF_Orchestration
KeyVault -.-> MAF_Orchestration
KeyVault -.-> MAF_Agents

MAF_Orchestration --> MAF_Agents
MAF_Agents --> Foundry
MAF_Agents --> MCP_Layer
MAF_Agents --> Knowledge

```

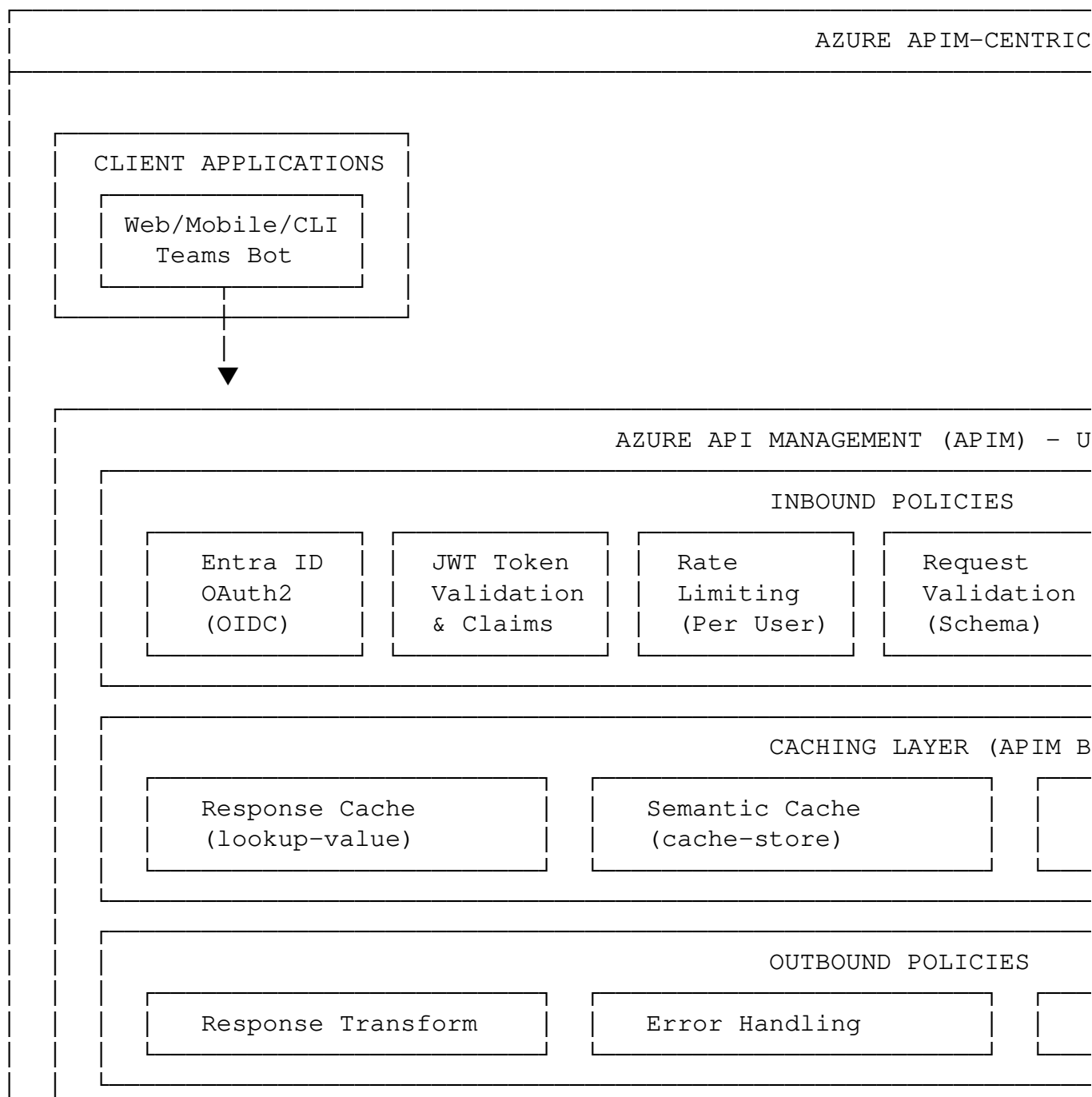
MAF_Orchestration --> Observability
MAF_Agents --> Observability

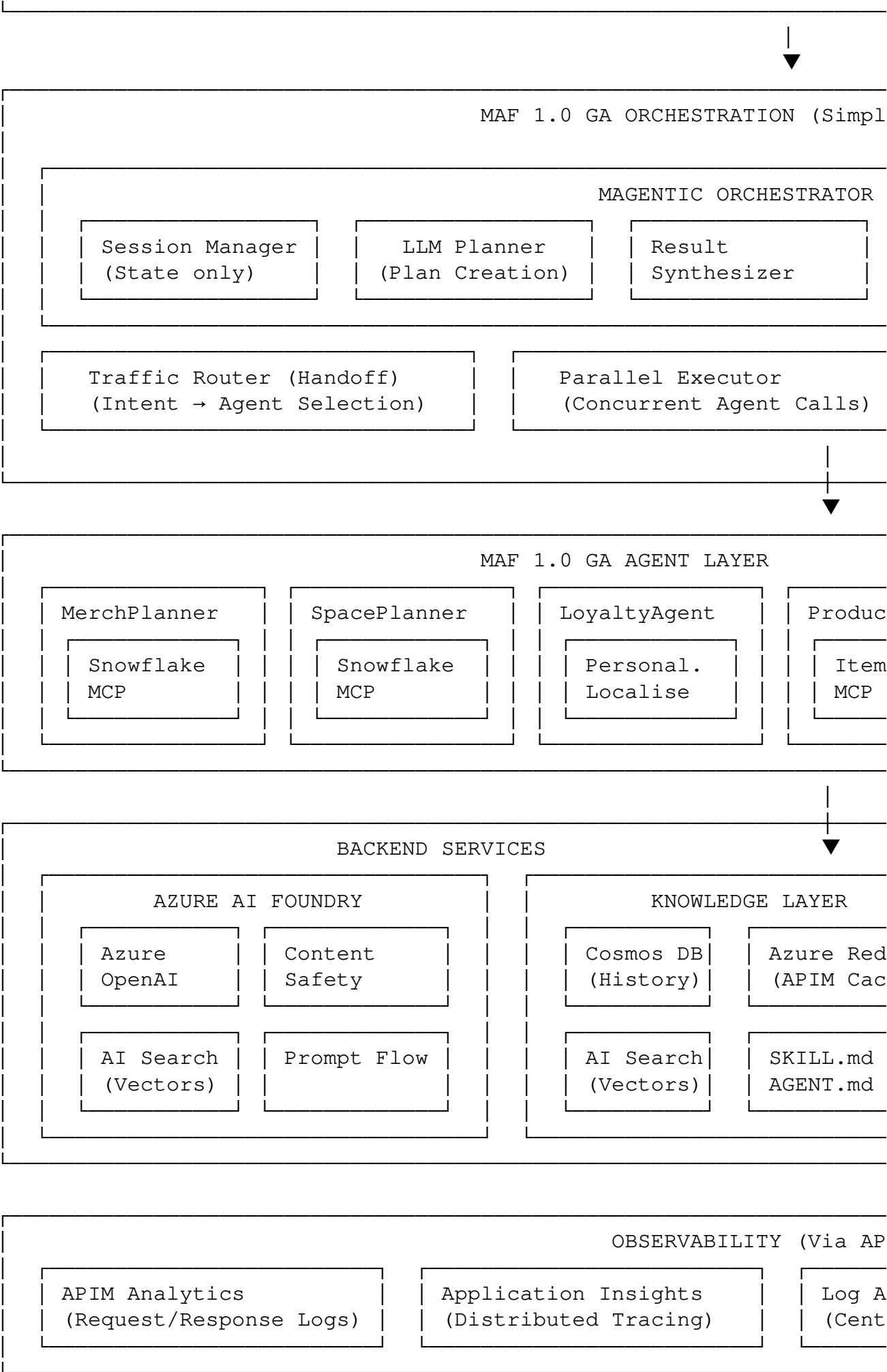
HITL --> Approval
Magentic --> ParallelExec
TrafficRouter --> MAF_Agents

Version 2: Azure APIM-Centric Architecture

This version uses Azure APIM to handle caching, validation, token management, and authentication, reducing custom middleware.

ASCII Diagram





LEGEND:

APIM HANDLES:	MAF 1.0 GA HANDLES:
✓ Authentication (Entra ID OAuth2)	✓ Task-Ledger Planning
✓ Token Validation (JWT Claims)	✓ Multi-Agent Orchestration
✓ Rate Limiting (Per User/Agent)	✓ Context Building & Memory
✓ Request/Response Caching	✓ Human-in-the-Loop Approval
✓ Schema Validation	✓ Result Synthesis
✓ PII Masking (Outbound)	✓ Agent Routing & Execution
✓ Logging & Tracing	✓ Parallel Execution

Mermaid Diagram

```
graph TD
    subgraph Clients ["👤 Client Applications"]
        WebUI["🌐 Web/Mobile/CLI/Teams"]
    end

    subgraph APIM ["🌐 Azure API Management - Unified Gateway"]
        subgraph InboundPolicies ["👉 Inbound Policies"]
            Auth["🔒 Entra ID OAuth2"]
            JWT["📄 JWT Validation"]
            RateLimit["🕒 Rate Limiting"]
            SchemaVal["✓ Schema Validation"]
            IPFilter["🚫 IP Filtering"]
            Quota["📊 Quota Management"]
        end

        subgraph CacheLayer ["⚡ Caching Layer"]
            ResponseCache["📦 Response Cache"]
            SemanticCache["🗄️ Semantic Cache"]
            LLMCache["🧠 LLM Response Cache"]
            ExternalCache["🔗 External Redis"]
        end

        subgraph OutboundPolicies ["👈 Outbound Policies"]
            Transform["🔄 Response Transform"]
            ErrorHandler["⚠️ Error Handling"]
            Logging["🔍 Logging & Tracing"]
            PIIMask["🚫 PII Masking"]
        end
    end

    subgraph MAF_Core ["⚙️ MAF 1.0 GA Core (No Auth Middleware Needed)"]
        subgraph Orchestration ["🌀 Orchestration Layer"]
            SessionMgr["🔑 Session Manager"]
            Planner["📅 LLM Planner"]
            Synthesizer["📝 Result Synthesizer"]
            ContextBuilder["🏗️ Context Builder"]
        end
    end
```

```

    AgentRouter["🔗 Agent Router"]
end

    TrafficRouter["🔗 Traffic Router"]
    ParallelExec["⚡ Parallel Executor"]
    HITL["👁 HITL via Service Bus"]
end

subgraph Agents[" MAF Agent Layer"]
    Agent1["📦 MerchPlanner"]
    Agent2["🏠 SpacePlanner"]
    Agent3["🎁 LoyaltyAgent"]
    Agent4["🔍 ProductsFinder"]
    Agent5["📋 CommercialSales"]
    Agent6["📈 CampaignAnalyst"]
end

subgraph Backend["🔧 Backend Services"]
    subgraph Foundry[" Azure AI Foundry"]
        OpenAI["Azure OpenAI"]
        Safety["Content Safety"]
        AISearch["AI Search"]
    end

    subgraph Knowledge["📖 Knowledge"]
        CosmosDB["Cosmos DB"]
        Redis["Redis<br/>(APIM Cache)"]
        VectorDB["Vector DB"]
    end

    subgraph MCP["🔌 MCP/A2A"]
        InternalMCP["Internal MCP"]
        ExternalA2A["External A2A"]
    end
end

subgraph Observability["📊 Observability"]
    APIMAnalytics["APIM Analytics"]
    AppInsights["App Insights"]
    LogAnalytics["Log Analytics"]
end

Clients --> Auth
Auth --> JWT --> RateLimit --> SchemaVal
SchemaVal --> ResponseCache
ResponseCache --> MAF_Core

MAF_Core --> Agents
Agents --> Backend

OutboundPolicies --> Observability
MAF_Core --> Observability

```

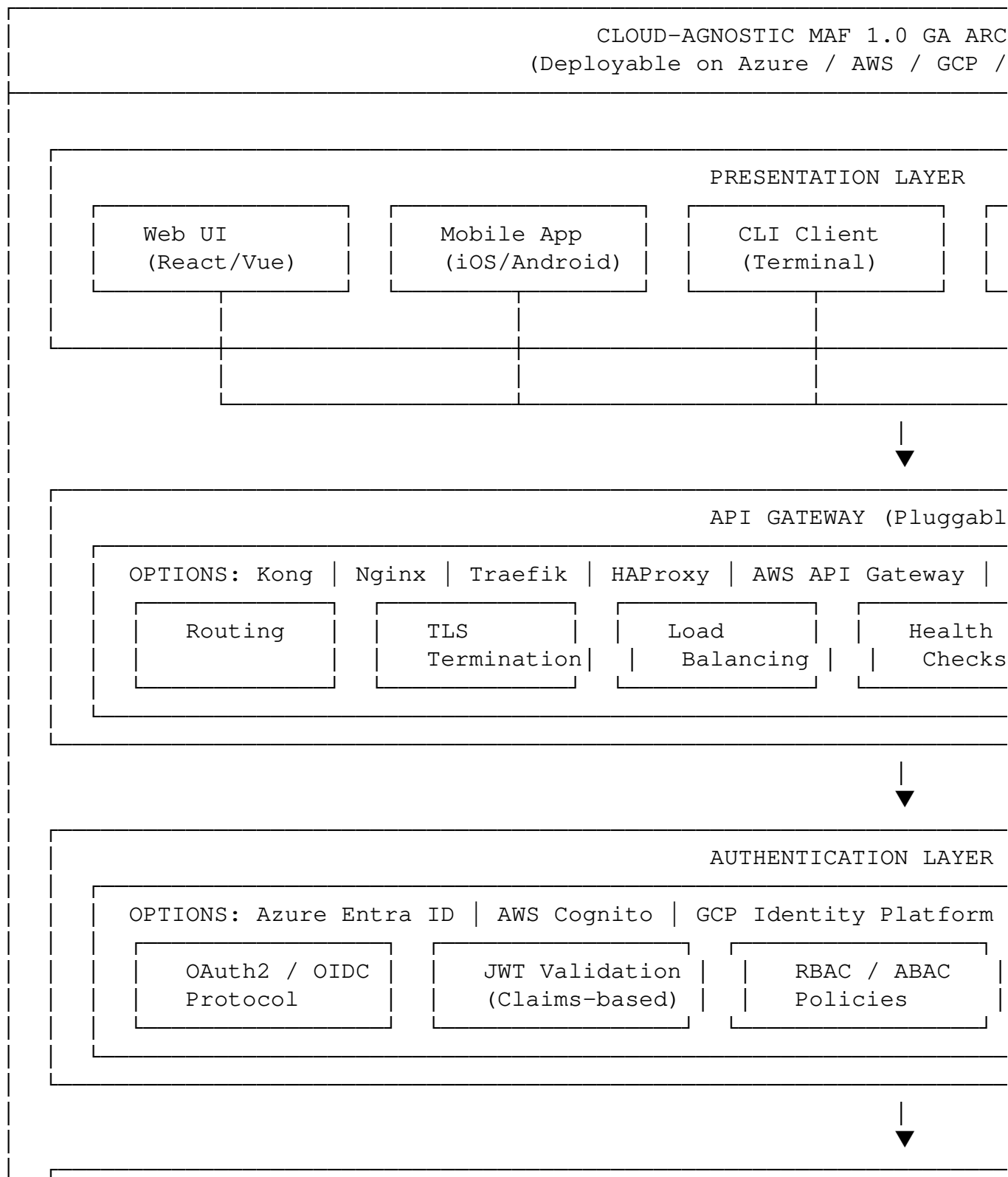


```
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style MAF_Core fill:#107c10,color:#fff
```

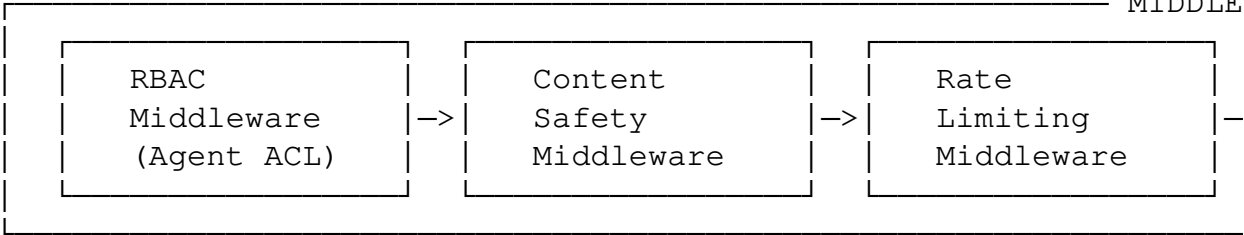
Version 3: Cloud-Agnostic MAF GA Architecture

This version is cloud-agnostic, supporting any cloud provider or on-premises deployment.

ASCII Diagram

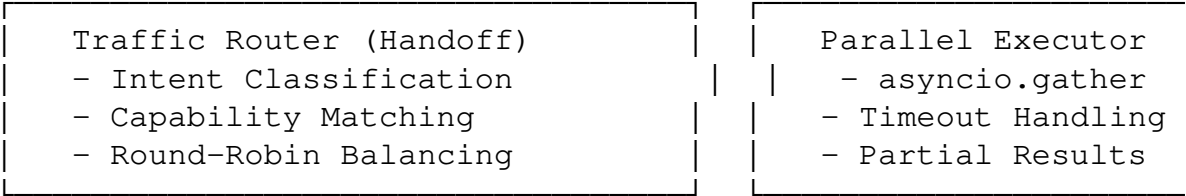
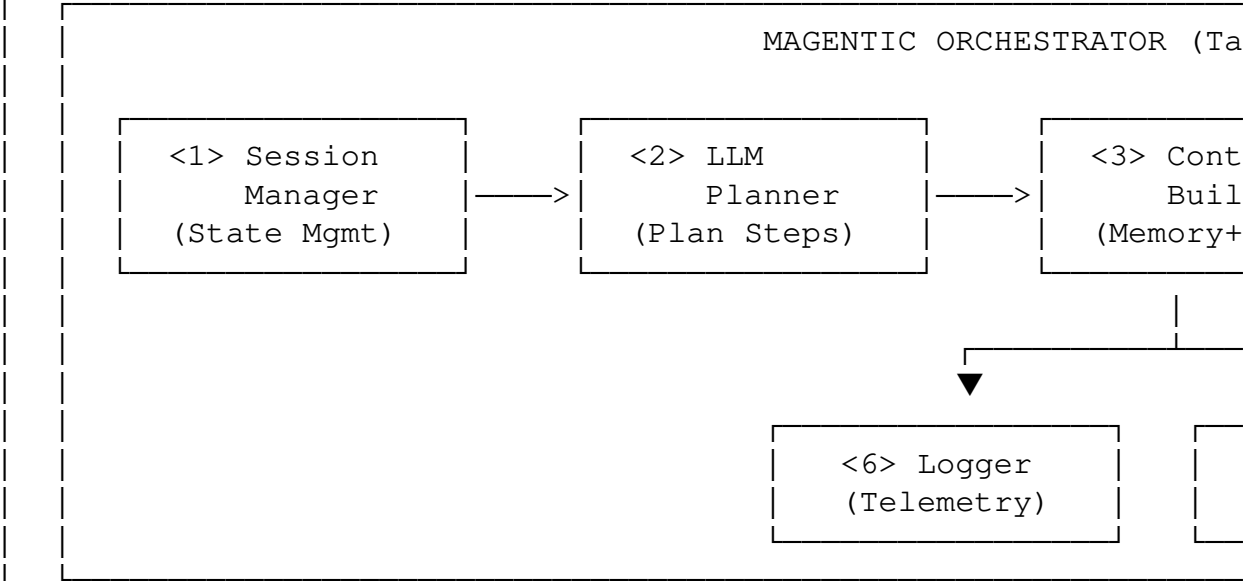


MIDDLE

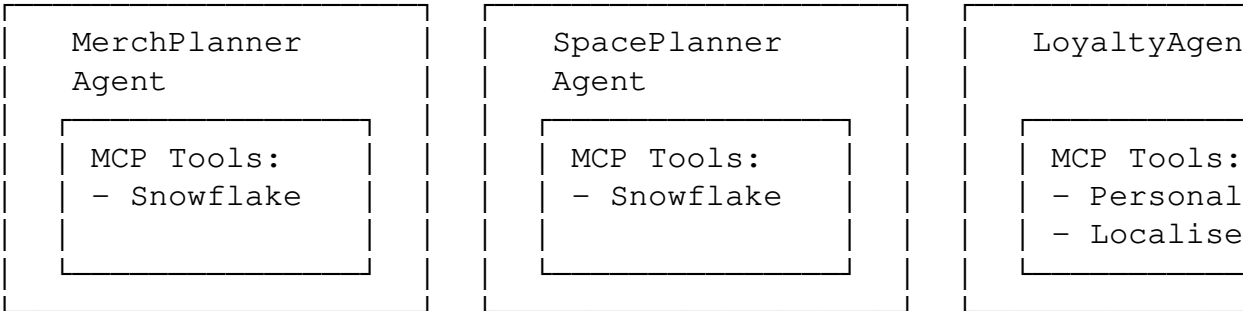


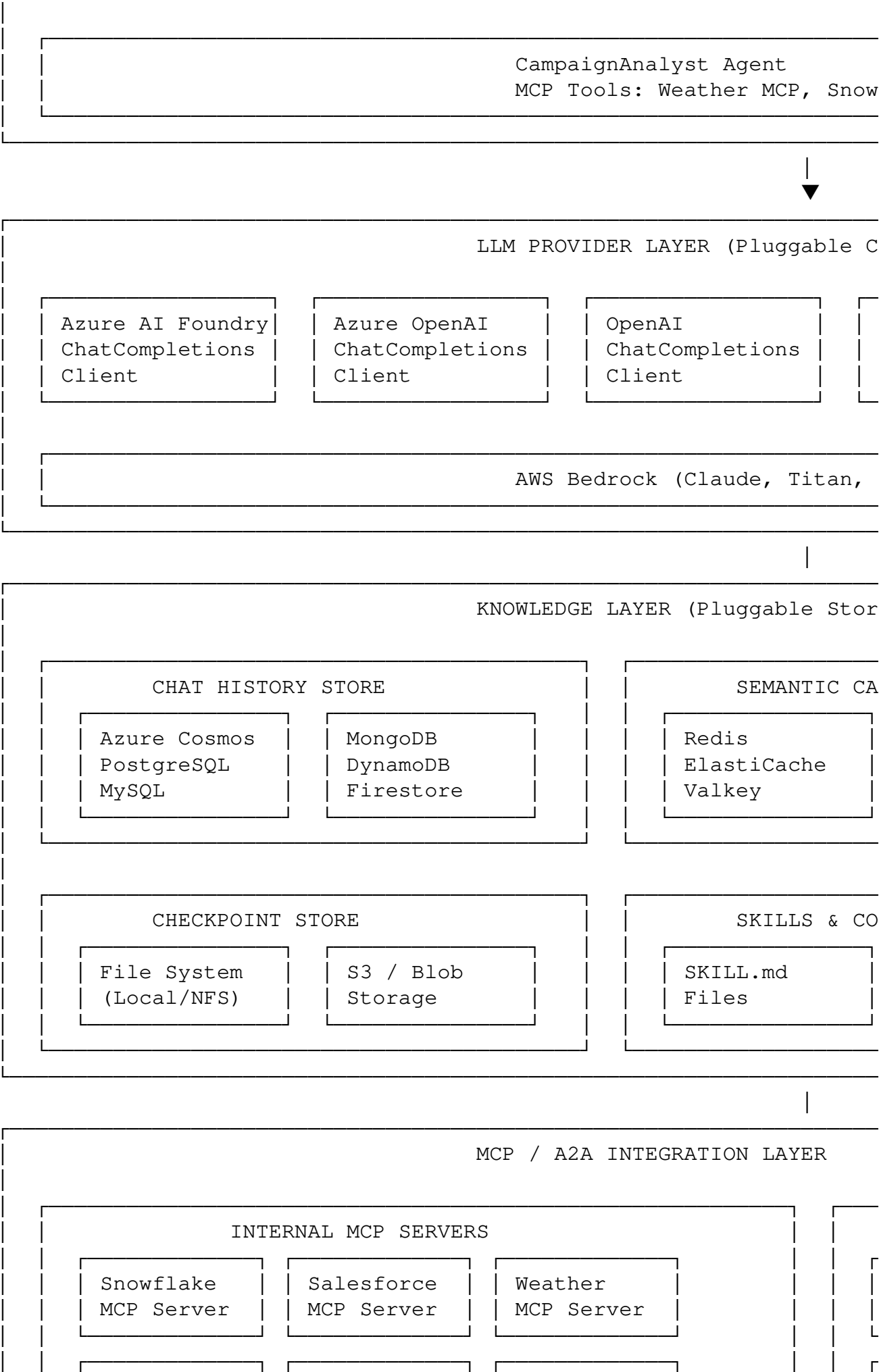
MAF 1.0 GA ORCHESTRATION LAY

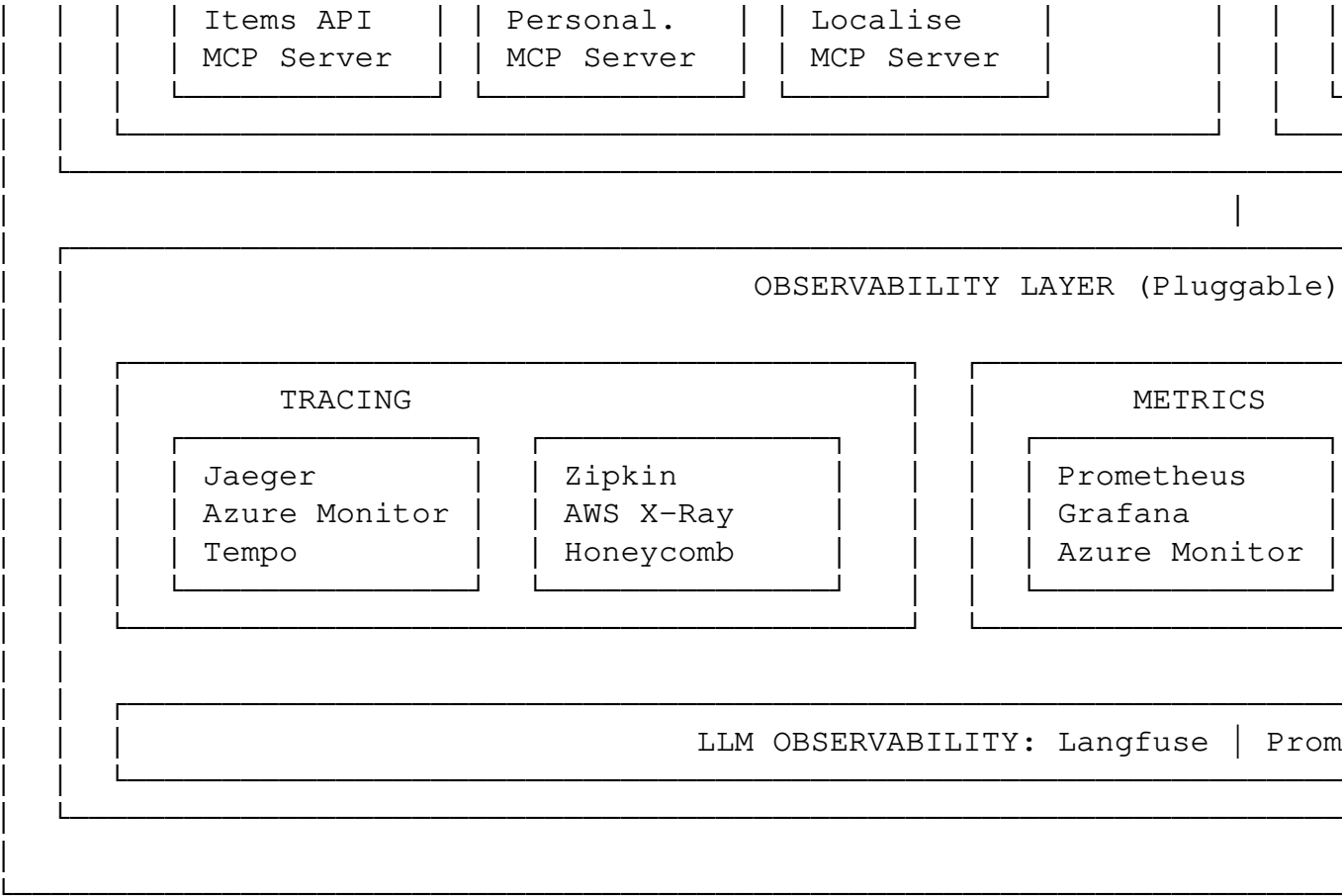
MAGENTIC ORCHESTRATOR (Ta



MAF 1.0 GA AGENT LAYER (Retail







Mermaid Diagram

```
flowchart TB
    subgraph Clients["👤 Client Applications (Any Platform)"]
        WebUI["🌐 Web UI"]
        Mobile["📱 Mobile"]
        CLI["💻 CLI"]
        Bot["💬 Chat Bot"]
    end

    subgraph Gateway["🌐 API Gateway (Pluggable)"]
        GatewayOptions["Kong | Nginx | Traefik | Envoy<br/>AWS API GW |"]
    end

    subgraph Auth["🔒 Auth Provider (Pluggable)"]
        AuthOptions["Azure Entra ID | AWS Cognito | GCP Identity<br/>Ok"]
    end

    subgraph FastAPI["⚡ FastAPI Service"]
        subgraph Middleware["🔄 Middleware Pipeline"]
            RBAC["❤️ RBAC"]
            Safety["🛡️ Content Safety"]
            RateLimit["🕒 Rate Limit"]
            Audit["📔 Audit Log"]
        end
    end

    subgraph MAF["⚙️ MAF 1.0 GA Framework"]
```

```

subgraph Orchestration["🌀 Orchestration Layer"]
    Magentic["Magentic Orchestrator<br/>(Task-Ledger)"]
    TrafficRouter["Traffic Router<br/>(Handoff)"]
    Parallel["Parallel Executor<br/>(Concurrent)"]
    HITL["HITL Manager"]
end

subgraph AgentLayer[" Agent Layer"]
    Agent1["MerchPlanner"]
    Agent2["SpacePlanner"]
    Agent3["LoyaltyAgent"]
    Agent4["ProductsFinder"]
    Agent5["CommercialSales"]
    Agent6["CampaignAnalyst"]
end

end

subgraph LLM[" LLM Providers (Pluggable)"]
    Azure["Azure Foundry/OpenAI"]
    OpenAI["OpenAI"]
    Anthropic["Anthropic Claude"]
    Gemini["Google Gemini"]
    Ollama["Ollama (Local)"]
    Bedrock["AWS Bedrock"]
end

end

subgraph Knowledge["📖 Knowledge Layer (Pluggable)"]
    subgraph ChatHistory["Chat History"]
        CosmosDB["Cosmos/Mongo/Postgres/DynamoDB"]
    end
    subgraph Cache["Semantic Cache"]
        Redis["Redis/Memcached/Valkey"]
    end
    subgraph VectorDB["Vector Store"]
        Vectors["AI Search/Pinecone/Weaviate/Qdrant"]
    end
    subgraph Checkpoint["Checkpoint"]
        FileStore["File/S3/Blob"]
    end
    Skills["SKILL.md / AGENT.md"]
end

end

subgraph MCP_A2A["🔗 MCP / A2A Layer"]
    subgraph InternalMCP["Internal MCP Servers"]
        Snowflake["Snowflake"]
        Salesforce["Salesforce"]
        Weather["Weather"]
        Items["Items API"]
    end
    subgraph ExternalA2A["External A2A Agents"]
        Pricing["Pricing"]
        Inventory["Inventory"]
        Fulfilment["Fulfilment"]
    end
end

```

```

    end
end

subgraph Observability["📊 Observability (Pluggable)"]
    subgraph Tracing["Tracing"]
        TracingOptions["Jaeger | Zipkin | Tempo | X-Ray"]
    end
    subgraph Metrics["Metrics"]
        MetricsOptions["Prometheus | Datadog | CloudWatch"]
    end
    subgraph Logging["Logging"]
        LoggingOptions["ELK | Loki | Splunk"]
    end
    end
    LLMObs["Langfuse | Promptlayer | Helicone"]
end

Clients --> Gateway
Gateway --> Auth
Auth --> FastAPI
FastAPI --> MAF
MAF --> LLM
MAF --> Knowledge
MAF --> MCP_A2A
MAF --> Observability

style MAF fill:#107c10,color:#fff
style LLM fill:#6b5b95,color:#fff
style MCP_A2A fill:#d64161,color:#fff

```

Summary Comparison

Component	V1: Azure-Native	V2: APIM-Centric	V3: Cloud-Agnostic
API Gateway	Azure APIM	Azure APIM (Extended)	Kong/Nginx/Traefik
Authentication	Azure Entra ID	Azure APIM + Entra ID	Any OIDC Provider
Secrets	Azure Key Vault	Azure Key Vault	Any Vault (HashiCorp, AWS Secrets)
LLM Provider	Azure AI Foundry	Azure AI Foundry	Any (OpenAI, Anthropic, etc.)
Middleware	MAF Custom	APIM Policies	MAF Custom
Caching	Azure Redis	APIM Built-in + Redis	Redis/Memcached
Vector DB	Azure AI Search	Azure AI Search	Pinecone/Weaviate/Qdrant
Chat History	Azure Cosmos DB	Azure Cosmos DB	MongoDB/PostgreSQL
Observability	Azure Monitor	APIM Analytics + Monitor	Prometheus/Jaeger/ELK
Complexity	Medium	Low (APIM handles more)	Medium
Vendor Lock-in	High (Azure)	High (Azure)	None

Cost	Higher (managed services)	Medium (APIM consolidation)	Variable
------	---------------------------	-----------------------------	----------

Deployment Recommendations

V1: Azure-Native

Best for: Enterprise environments with existing Azure investment, compliance requirements, and need for managed services.

V2: APIM-Centric

Best for: Teams wanting to minimize custom middleware, leverage APIM policies for security/caching, and simplify operations.

V3: Cloud-Agnostic

Best for: Multi-cloud strategies, on-premises requirements, open-source preference, or avoiding vendor lock-in.